1 GCR-4 Method and the 4DVisualizer Environment

The GCR-4 method provides a geometric reinterpretation of robot motion in six degrees of freedom. Instead of projecting the data onto statistical axes, it embeds each configuration into a four-dimensional barycentric space. This transformation preserves the continuity and structure of the robot's trajectory, allowing the motion to be examined as a coherent curve rather than a cloud of unrelated points. The accompanying 4DVisualizer environment serves as the medium through which this curve becomes visible and interpretable.

Robot motion is typically represented by a sequence of positions and orientations. Although complete, this representation is difficult to visualize directly. GCR-4 addresses this challenge by mapping each configuration into a four-dimensional simplex. Three coordinates describe the barycentric position within the simplex, while the fourth acts as the complement that closes the system. The transformation ensures that every point lies within a normalized geometric domain, making the resulting trajectory suitable for visual inspection. Sudden disturbances in orientation or position appear as visible irregularities in the curve, revealing anomalies that may remain hidden in raw numerical data.

Principal Component Analysis approaches the same problem from a fundamentally different perspective. By searching for directions of maximal variance, PCA compresses the motion into a linear subspace. When applied to smooth robot trajectories, this often results in a nearly straight line in the principal component domain. Any irregular point becomes an outlier relative to this line, but the underlying shape of the motion is lost. The distinction is conceptual: PCA simplifies the motion into a linear structure, while GCR-4 preserves its nonlinear geometry. In the context of robot motion analysis, this difference is significant, because the robot does not move along a statistical axis but along a continuous path in configuration space.

The 4DVisualizer environment is designed to display the barycentric curve produced by GCR-4. Its purpose is not to perform dimensionality reduction but to reveal the structure that the method uncovers. Within this environment, the curve can be examined as a coherent geometric object. Variations in curvature, density, or direction correspond directly to variations in the robot's motion. Anomalies stand out naturally, not because they are statistically distant, but because they disrupt the continuity of the curve.

The tool therefore acts as a bridge between abstract geometric transformation and practical interpretation.

The combination of GCR-4 and 4DVisualizer is well suited to industrial and research settings. It can be used to compare repeated motion cycles, diagnose calibration issues, monitor long-term drift, or study the structure of complex trajectories. Because the method preserves geometric relationships, it provides a reliable basis for detecting subtle deviations that might otherwise remain hidden in raw 6D data.

In summary, the GCR-4 method transforms robot motion into a form that is both compact and geometrically meaningful. When paired with the 4DVisualizer environment, it enables a style of analysis that emphasizes continuity, structure, and interpretability. In contrast to linear statistical approaches, it presents the robot's trajectory as a curve rather than a projection, making it possible to observe the motion as a coherent geometric phenomenon.